Total No. of Pages: 2 Roll No.....

VITH SEMESTER B.Tech.(Computer Engineering)

Supplementary END SEMESTEREXAMINATION

(September 2019)

CO 304 ARTIFICIAL INTELLIGENCE

Paper Code

Time: 3 Hours Max. Marks: 50

Note: Attempt total 5 questions. Q1 is compulsory

Q1. Answer following to the point.

a. Give an example of intelligent and Non intelligent program for Tic Tae Toe Problem.
b. Give any five project characteristics to analyze a problem domain
c. Explain the concept of knowledge. What are the different ways to represent Knowledge?
d. State Bayes theorem and support with example for statistical reasoning.
e. What are the different techniques for non monotonic reasoning?
f. Explain rule based expert system

(5 X 2)

Q2.(a) Explain A* procedure to reach goal node from starting initial position.

(b) Write down search procedure for And/or graph

[5,5]

Q3. (i) Explain the concept of game tree with suitable example
(ii) Explain steps to solve crypt arithmetic problems

[5,5]

Q4. Consider the following sentences:

a. Every child loves dog.
b. All dogs have fur skin or short hair.
c. Boby is a dog
d. Mina does not love dog with short hair
e. Rekha loves Boby
f. Mina does not love dog

I. Represent theses sentences as wff.
II. Using backward chaining prove that Mina is not child.
III. Convert them to clause form indicating the law used.
IV. Explaining steps of resolution; prove that Boby has fur skin.

[3,3,4,4]

Q5. Explain the different concepts used in Semantic Net using suitable example. Also write the corresponding inference procedure.How semantic nets are extended to semantic Frames. What are its merit and demerits over semantic nets.

[5,5]

Q6. (a)) What are the steps to build Expert system such as Medical diagnostic system. Explain its architecture.

(b) Explain the concept of Fuzzy set with example. What operations can be performed on these set. Illustrate with example.

[6,4]

Q7. Write short notes on following (Any two)

(a) Conceptual Dependency

(b) Different phases of Natural Language processing

(c) Different Learning algorithm.

(d) Non monotonic reasoning

[5,5]